

1. **Understand student reports**

AI reads what the student types and understands what problem they are experiencing.

2. **Automatically categorise problems**

It identifies whether the issue is:

- Maintenance
- Wi-Fi/IT
- Plumbing
- Electricity
- Security
- Cleaning
- Laundry
- Noise
- Furniture
- Other

3. **Determine urgency**

AI decides whether an issue is **Low, Medium, High, or Critical** based on its potential impact.

4. **Automatically route the report**

AI determines which department should handle the issue, such as Maintenance, IT, Security, or Residence Management.

5. **Detect duplicate reports**

If 20 students report the same broken Wi-Fi connection, AI can recognise that these may be the same underlying problem rather than 20 separate problems.

6. **Identify larger problems**

AI can analyse multiple reports and identify patterns, such as:

"Several students in Block B are experiencing water problems."

7. **Analyse uploaded photos**

If implemented, students can upload a picture of the problem and AI can help identify what it shows—for example, a damaged pipe, broken furniture, or a damaged facility.

8. **Generate a short summary for staff**

Instead of staff reading a long student complaint, AI can summarise it:

Problem: Broken washing machine

Location: Block C

Priority: Medium

9. **Recommend actions**

AI can suggest what department or type of response may be required.

10. **Monitor recurring problems**

AI can identify issues that repeatedly occur in the same building, room, or facility.

11. **Generate management insights**

AI can tell management things like:

"Wi-Fi complaints increased by 40% this week."

12. **Identify trends**

AI can analyse reports over time to determine which problems are becoming more common.

13. Predict potential maintenance problems

In a future version, AI could use historical data to predict which equipment or facilities may require maintenance.

14. Create automated reports

AI could generate weekly or monthly summaries for residence management, showing the most common problems, unresolved issues, response times, and trends.

15. Power an AI chatbot

Students could ask:

"How do I report a broken washing machine?"

or:

"What is the status of my maintenance request?"

16. Answer common student questions

AI could provide information about residence rules, facilities, reporting procedures, Wi-Fi troubleshooting, and other frequently asked questions.

17. Provide automatic updates

AI could help generate clear notifications when the status of a student's report changes.

18. Identify potentially serious safety issues

AI can flag reports that appear to involve serious security or safety concerns so they receive immediate attention.

19. Analyse student feedback

AI could analyse comments and feedback to identify common frustrations and areas where the residence could improve.

20. Measure accommodation performance

AI could analyse how quickly problems are being resolved and identify departments or problem types with slow response times.

The main AI workflow

Student submits a report → AI understands it → categorises it → determines priority → detects duplicates → sends it to the right department → monitors patterns → provides insights to management.

For the **first prototype**, I would recommend focusing on **5 core AI features** so the project stays realistic:

1. Problem classification

2. Priority detection

3. Automatic department assignment

4. Duplicate/problem pattern detection

5. AI-generated management insights